

Day 05 – Linux Troubleshooting Drill: CPU, Memory, and Logs

A) Environment basics

- `uname -a` → shows kernel, system architecture, and OS details
- `lsb_release -a` → shows Linux distribution name, version, and codename
- `cat /etc/os-release` → shows OS name and version from system file

```
ubuntu@ip-172-31-23-127:~$ uname -a
Linux ip-172-31-23-127 6.14.0-1018-aws #18~24.04.1-Ubuntu SMP Mon Nov 24 19:46:27 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
ubuntu@ip-172-31-23-127:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:   Ubuntu 24.04.3 LTS
Release:      24.04
Codename:     noble
ubuntu@ip-172-31-23-127:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.3 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.3 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
ubuntu@ip-172-31-23-127:~$ |
```

B) Filesystem sanity

- temporary directory was created and a file was copied into it.
- This confirms the filesystem is accessible and read/write operations are working

```
ubuntu@ip-172-31-23-127:~$ mkdir -p /tmp/runbook-demo
ubuntu@ip-172-31-23-127:~$ cd /tmp/
ubuntu@ip-172-31-23-127:/tmp$ ls
runbook-demo
snap-private-tmp                                systemd-private-b8b1d9fc011e4a4695e5cd75a05954d5-polkit.service-HS60Ju
systemd-private-b8b1d9fc011e4a4695e5cd75a05954d5-ModemManager.service-Q9CtAH  systemd-private-b8b1d9fc011e4a4695e5cd75a05954d5-systemd-logind.service-Bdk3Yt
systemd-private-b8b1d9fc011e4a4695e5cd75a05954d5-chrony.service-G80e2e         systemd-private-b8b1d9fc011e4a4695e5cd75a05954d5-systemd-resolved.service-p2D3MN
ubuntu@ip-172-31-23-127:/tmp$ cp /etc/hosts /tmp/runbook-demo/hosts-copy
ubuntu@ip-172-31-23-127:/tmp$ cd runbook-demo/
ubuntu@ip-172-31-23-127:/tmp/runbook-demo$ ls
hosts-copy
ubuntu@ip-172-31-23-127:/tmp/runbook-demo$ ls -l /tmp/runbook-demo
total 4
-rw-r--r-- 1 ubuntu ubuntu 221 Feb  9 16:31 hosts-copy
ubuntu@ip-172-31-23-127:/tmp/runbook-demo$ |
```

c) CPU / Memory

top : Shows real-time CPU & memory usage for all running processes

Htop : An interactive, user-friendly version of top with colors, mouse support, and easy process killing.

ps -o pid,pcpu,pmem,comm -p \$(pidof sshd) : Gets all sshd process IDs and shows their PID, CPU %, Memory %, and command name in one shot

free -h : shows total, used, free, and available RAM (human-readable)

```
ubuntu@ip-172-31-23-127: /tmp/runbook-demo$ top
top - 16:38:57 up 28 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 113 total, 1 running, 112 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 4.5 sy, 0.0 ni, 95.5 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 914.2 total, 223.4 free, 401.0 used, 447.9 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 513.2 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   2212    1364   9628  S   0.0   1.4   0:01.17 systemd
    2 root        20   0      0      0      0  S   0.0   0.0   0:00.00 kthreadd
    3 root        20   0      0      0      0  S   0.0   0.0   0:00.00 pool_workqueue_release
    4 root        0  20      0      0      0  T   0.0   0.0   0:00.00 /usr/bin/python3

ubuntu@ip-172-31-23-127:~$ ps -o pid,pcpu,pmem,comm -p $(pidof sshd)
  PID %CPU %MEM    COMMAND
 1257  0.0   0.8    sshd
 2491  0.0   1.1    sshd
 2561  0.0   0.7    sshd
ubuntu@ip-172-31-23-127:~$

2561  0.0   0.7    sshd
ubuntu@ip-172-31-23-127:~$ free -h
             total        used        free      shared  buff/cache   available
Mem:        914Mi       388Mi       219Mi         2.7Mi        465Mi       525Mi
```

D) Disk / IO

- **df -h** → Shows disk space usage of all mounted filesystems in human-readable format
- **du -sh /var/log** → Shows total disk space used by the /var/log directory
- **iostat** → Displays CPU usage and disk read/write I/O statistics
- **vmstat** → Shows memory, swap, CPU, and I/O system activity
- **dstat** → Shows combined real-time CPU, disk, memory, and network I/O stats

```
ubuntu@ip-172-31-23-127:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        19G   2.5G   16G   14% /
tmpfs            458M   0    458M   0% /dev/shm
tmpfs            183M  896K   182M   1% /run
tmpfs            5.0M   0    5.0M   0% /run/lock
efivarfs         128K   3.8K   120K   4% /sys/firmware/efi/efivars
/dev/nvme0n1p16  881M   89M   730M  11% /boot
/dev/nvme0n1p15  105M   6.2M   99M    6% /boot/efi
tmpfs            92M   12K    92M   1% /run/user/1000
ubuntu@ip-172-31-23-127:~$
```

```
ubuntu@ip-172-31-23-127:~$
ubuntu@ip-172-31-23-127:~$ sudo du -sh /var/log
56M    /var/log
ubuntu@ip-172-31-23-127:~$
```

```
ubuntu@ip-172-31-23-127:~$
ubuntu@ip-172-31-23-127:~$ iostat
linux 6.14.0-1018-aws (ip-172-31-23-127)      02/09/26      _x86_64_      (2 CPU)

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
            0.35    0.00    0.22    0.06    0.02   99.35

Device            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd
loop0              0.06      1.54         0.00         0.00       4271         0         0
loop1              0.01      0.12         0.00         0.00        345         0         0
loop2              0.02      0.39         0.00         0.00       1082         0         0
loop3              0.02      0.12         0.00         0.00        346         0         0
loop4              0.02      0.39         0.00         0.00       1097         0         0
loop5              0.22      8.14         0.00         0.00      22621         0         0
loop6              0.00      0.00         0.00         0.00         10         0         0
nvme0n1            4.61     172.58        17.51         0.00     479503     48646         0

ubuntu@ip-172-31-23-127:~$ vmstat
procs-----memory----- --swap--  -----io----- -system--  -----cpu-----
 r b swpd free buff cache  si so  bi bo  in cs us sy id wa st gu
 2 0   0 231324 14784 463968    0 0  183  17 124  0 0 0 99 0 0 0
ubuntu@ip-172-31-23-127:~$ dstat
```

E) Network

- `ss -tulpn` → Shows listening TCP/UDP ports and the processes using them
- `netstat -tulpn` → Displays listening network ports and associated services
- `curl -I <service-endpoint>` → Checks if a service endpoint is reachable via HTTP response headers
- `ping <host>` → Tests basic network connectivity and latency to a host

```
ubuntu@ip-172-31-23-127:~$ ss -tulpn | grep sshd
ubuntu@ip-172-31-23-127:~$ sudo ss -tulpn | grep sshd
tcp    LISTEN 0      4096      0.0.0.0:22      0.0.0.0:*      users:((("sshd",pid=1257,fd=3),("systemd",pid=1,fd=90)))
tcp    LISTEN 0      4096      [::]:22        [::]:*         users:((("sshd",pid=1257,fd=4),("systemd",pid=1,fd=91)))
ubuntu@ip-172-31-23-127:~$
```

```
ubuntu@ip-172-31-23-127:~$
ubuntu@ip-172-31-23-127:~$ nc -zv localhost 22
Connection to localhost (127.0.0.1) 22 port [tcp/ssh] succeeded!
ubuntu@ip-172-31-23-127:~$
ubuntu@ip-172-31-23-127:~$
```

F) Logs

- `journalctl -u <service> -n 50` → Shows the **last 50 log entries** for a specific systemd service
- `tail -n 50 /var/log/<file>.log` → Displays the **last 50 lines** of a log file in real time context

All checks show the system is **working fine**.

The server details are correct, files and folders can be created, CPU and memory usage look normal, disk space is not full, network services are reachable, and logs do not show any recent errors. The selected service is **running and stable**.

- **If this gets worse (next steps)**
- **Restart the service** and watch CPU, memory, and logs.
- **Check logs again** with more detail to find errors.
- **Run deeper checks** (like detailed I/O or process tracing) to find the root problem.